

Greenhouse Gasses Emissions in World's Economy Report

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Introduction

All data manipulations, visualizations, and textual insights were written within a single R Markdown file. R Markdown allowed me for seamless integration of R code, markdown, and LaTeX languages to ensure reproducibility, clarity, and transparency in how the analysis was conducted. The raw data was processed using various R functions, including filtering, grouping and derivation of plotted indicators. The `dplyr` and `tidyr` libraries were used to clean, transform, and aggregate the data where necessary. While visualizations were generated using `ggplot2` and `treemapify` packages.

- In chart 1 the year on year percentage change was calculated for EU27, Euro area (EA) and World Greenhouse Gasses (GHG) totals. The Euro area GHG emission was derived by summing total for countries belonging to EA. I presented the data on a line chart with visible data points to easily follow progression over time.
- In chart 2 I extracted data on income categories and populations totals from the publicly available World Bank database. I performed data cleaning (re-coded country codes and removed incomplete data) to synchronize data from EDGAR group and World Bank. I calculated the CO₂ emissions per capital within each of the income groups for year 2023 and presented in a bar chart for easy comparison between aggregates.
- In chart 3 I collected data representing mapping between country and continent. I extracted only individual countries' data. I calculated the percentage of Total GHG emission for each of 7 continent as well as individual country. I used nested tree map to visualize the proportions of total for each continent as well as each country.

The report's source code can be found on my github website here.

Insights report

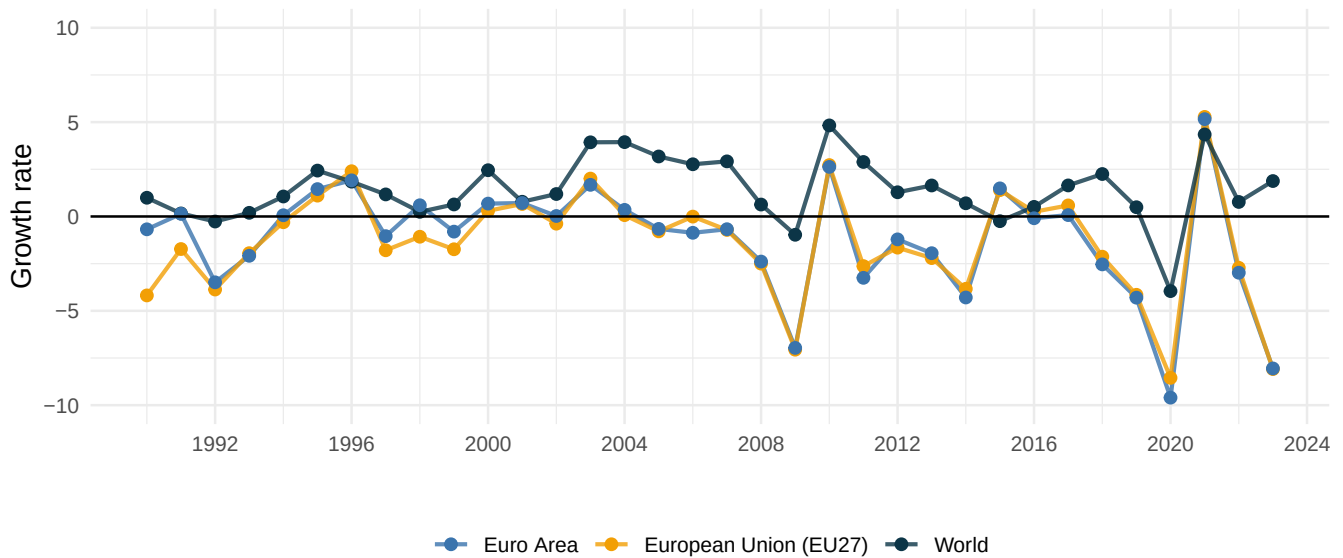
This report was prepared based on the EDGAR GHG Emissions database coming from 2024 'GHG emissions of all world countries' Publication (Crippa and Pekar 2024) ([report link](#))

Evolution of GHG growth in the euro area, European Union (EU27) and worldwide

- In the European Union (EU27) and Euro Area, there was a noticeable 8.1% (or repetitively 261 Mt CO_{2-eq} and 196 Mt CO_{2-eq}) decrease in total GHG emissions in 2023 compared to 2022. On the other hand, the World GHG emission increased by 1.8% (994 Mt CO_{2-eq}).
- From 1990 to 2023 the EU27 and Euro Area countries showed steady decline in GHG emissions with -1.2% compounded annual growth rate (CAGR) while Total World GHG emissions continue to raise with CAGR equal to 1.5%.

Greenhouse Gasses Emission – Year on year growth rate

percentage change in total GHG emission

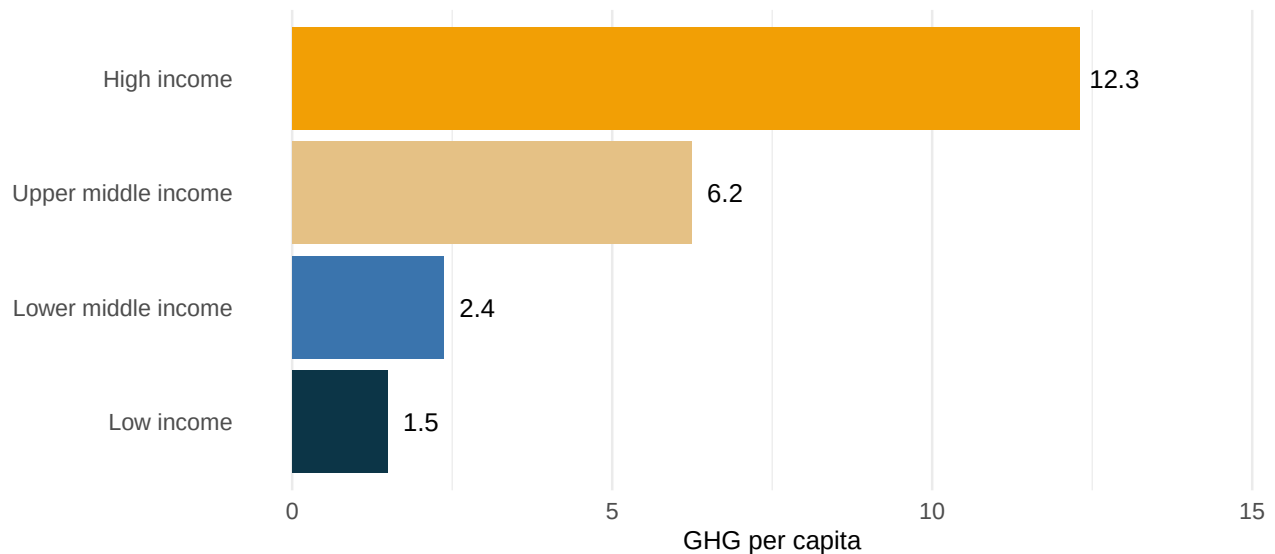


Comparison of countries' GHG emissions per capita aggregated according to the World Bank income groups

- On average high-income countries emit around 11.8 t of CO_{2-eq} per capita whereas low-income countries emit almost 8 times less tonnes of CO_{2-eq} per capita with average of 1.5 t CO_{2-eq} per capita which highlights disproportional emissions responsibility of wealthier nations.
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Greenhouse Gasses Emission per capita – 2023 income groups comparison

Tonnes of CO₂-eq per capita per year



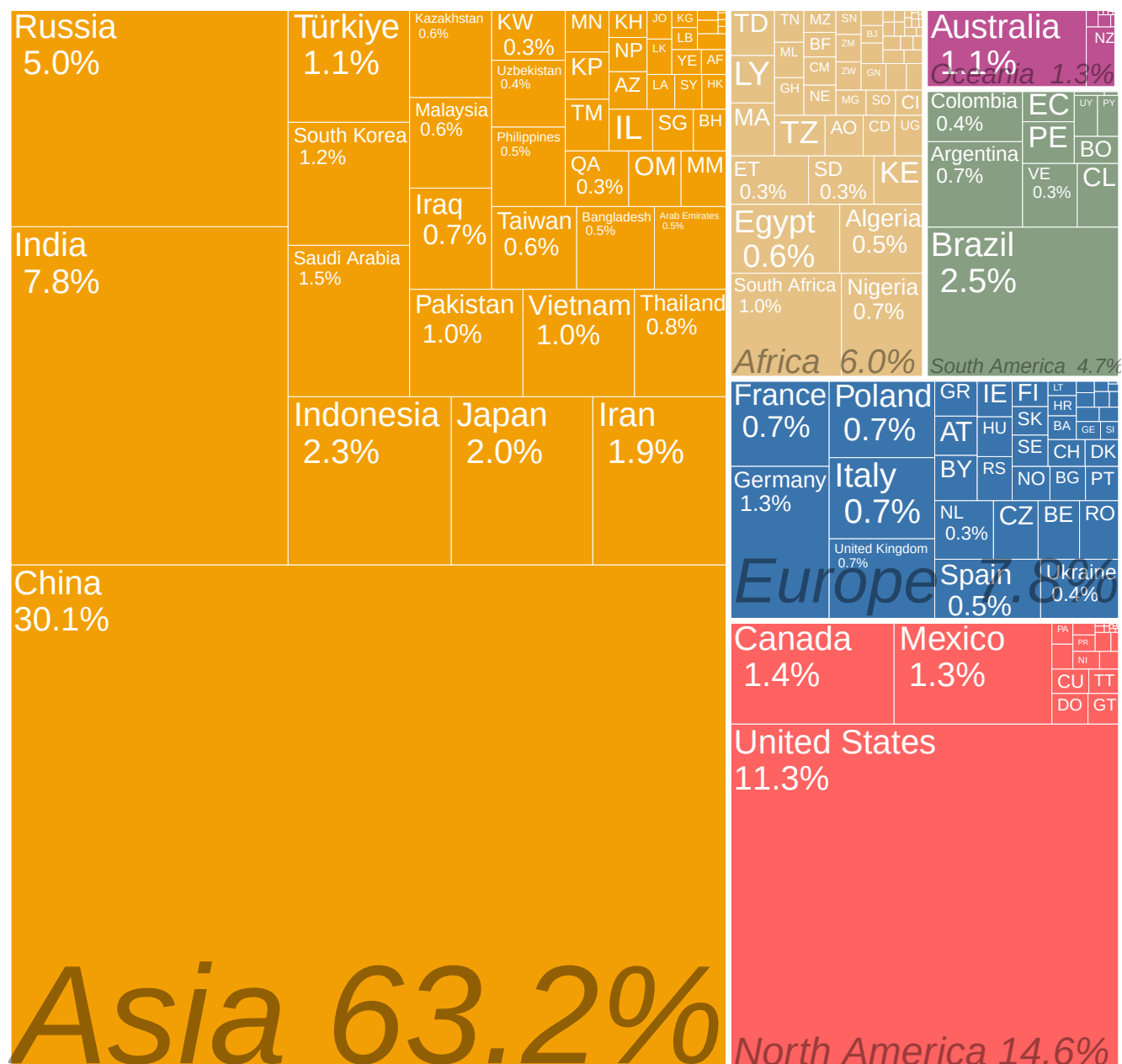
Note.(1) Individual countries were classified based on the latest classification provided by World Bank (<https://datatopics.worldbank.org/world-development-indicators/the-world-by-income-and-region.html>).
 (2) The GHG per capita was calculated using population totals from 2023 provided by World Bank (<https://data.worldbank.org/indicator/SP.POP.T>).

Contribution of individual country and continent GHG emissions to total world GHG emissions

- China, the United States, India, Russia and Brazil were the world's largest GHG emitters in 2023. Together they account for 49.8% of global population and 62.7% of global GHG emissions.
- Asia is responsible for the largest share of emissions, mainly due to rapid industrialization in countries like China and India. In African continent although the population is second largest the industrialization is on an incomparable lower level and in total accounts for 6
- In terms of contribution to the Europe GHG emissions in 2023, Germany remained the largest emitter, followed by France, Italy, Poland and Spain.

Contribution of continent and individual country GHG emissions to total World GHG emissions

Proportion of Total GHG emission in %



References

Crippa, Guizzardi, M., and F. Pekar. 2024. "GHG Emissions of All World Countries." *Publications Office of the European Union, Luxembourg*. <https://doi.org/10.2760/4002897>.